

$$\begin{aligned} & \Rightarrow f(a-b) \in I \\ & \Rightarrow a-b \in f^{-1}(I) \end{aligned}$$

$$\therefore f(a) \in I \text{ and } f(\lambda) \in R'$$

$$\therefore f(\lambda)f(a) \in I$$

$$\therefore f(\lambda a) \in I$$

$$\therefore \lambda a \in f^{-1}(I)$$

2. $R \rightarrow$ integral domain.

$$R[x]$$

$$R[x] \rightarrow \text{I.D.}$$

$$\{a_0 + a_1x + \dots + a_nx^n \mid a_i \in R, n \in \mathbb{N}\}$$

p.t. $R[x]$ is I.D. by contr.
(Assume $a_n b_m = 0$)

$$f = a_0 + \dots + a_n x^n \rightarrow \text{unit in } R[x]$$

$$\exists g \in R[x] \text{ s.t. } fg = 1.$$

$$\Rightarrow df + dg(g) = 0 \quad (dg1 = 0)$$

$$\Rightarrow df dg f = 0$$

$$\therefore f = a_0$$

$$\therefore U(R[x]) = U(R).$$

Claim: $f = \sum_{i=0}^n a_i x^i \in R[x]$ unit iff a_0 is unit in R

Let a_0 is unit;

$$f = a_0 + a_1 x + a_2 x^2 + \dots$$

$$\text{you want } g = b_0 + b_1 x + b_2 x^2 + \dots$$

$$\text{s.t. } fg = 1$$

$$\therefore a_0 b_0 = 1 \Rightarrow b_0 = a_0^{-1} \quad (\because a_0 \text{ is invertible})$$

$$\therefore a_0 b_1 + b_0 a_1 = 0 \Rightarrow a_0 b_1 + a_0^{-1} a_1 = 0 \Rightarrow b_1 = -a_0^{-2} a_1$$

$$\therefore a_0 b_2 + b_0 a_2 + a_1 b_1 = 0$$

5. $K[[X]]$; $K = \text{field}$

$a_0 = 0$ is not a unit $\Rightarrow 0 + x \text{ or } \dots$ is not invertible
 $\downarrow$
 field

Take I which is maximal ideal.

Claim: $I = (X)$

$f \in I$

$\downarrow$
 maximal ideal

$\Rightarrow f$ is not a unit

$[I \neq R]$

all

or

$a_0^{-1} = (a_0^{-1} + 0x + \dots)$

$I = R$

$f = a_1x + a_2x^2 + \dots$ [$\because a_0 = \text{non-unit}$]

$\therefore f = x(a_1 + a_2x + \dots)$

$\therefore f = x \cdot g$; $g \in R$

$\therefore f \in (x) \therefore f \in (x) \rightarrow \text{true}$

$\therefore I \subseteq (x)$

$\therefore I = (x)$

$\therefore R$ ideal is unique

$\Downarrow$

$R \rightarrow \text{ring}$

$I \rightarrow \text{ideal}$

$\frac{R}{I}$

$\downarrow$

Ideal of R/I is of form J/I where $I \subseteq J$.

$\frac{\mathbb{Z}}{4\mathbb{Z}}$	$\frac{2\mathbb{Z}}{4\mathbb{Z}}, \frac{2\mathbb{Z}}{4\mathbb{Z}}, \frac{4\mathbb{Z}}{4\mathbb{Z}} = 0$
$4\mathbb{Z}, 2\mathbb{Z}, \mathbb{Z}$	$\times$

R is a field iff (0) is a maximal ideal

$\frac{\mathbb{Z}}{4\mathbb{Z}} \not\cong \frac{\mathbb{Z}}{2\mathbb{Z}}$ $\rightarrow$ field $\because$ it is maximal ideal
iff R/I is field

$f: R \rightarrow R'$

$I \trianglelefteq R$

$f^{-1}(I) \trianglelefteq R$

$R \rightarrow R'$

$R' \rightarrow R'$

I

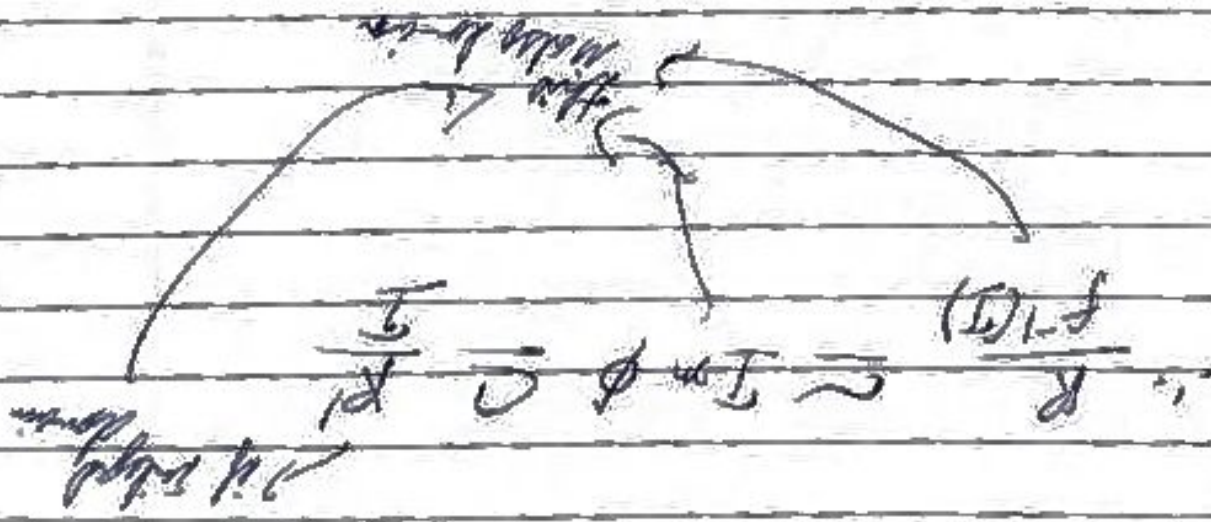
$a \mapsto f(a) \pmod{(f(I) + I)}$

$a > b \Rightarrow f(a) > f(b)$
 $\exists \text{ s.t. } f(a) = f(b) \Rightarrow a = b$
 $f(a) > f(b) \Rightarrow a > b$
 $f(a) = f(b) \Rightarrow a = b$
 $f(a) < f(b) \Rightarrow a < b$

1. $f: \mathbb{R} \rightarrow \mathbb{R}$ continuous

$f^{-1}(y) = \{x \mid f(x) = y\}$ is not a subset of $\mathbb{R}$

$f(x) = x$
 $f^{-1}(y) = y$
 $f^{-1}(y) = \{y\}$



$f^{-1}(y) = \{x \mid f(x) = y\}$
 $f(x) = y$
 $f^{-1}(y) = \{y\}$

$$\Rightarrow |f(x) - f(y)| < \epsilon \Rightarrow f \text{ is Cts.}$$

$$\therefore f(x) = \alpha x$$

$$(17) \dots$$

$$\frac{1}{x} \leq \frac{1}{y} \leq \frac{1}{x} \Rightarrow \frac{1}{x} \leq \frac{1}{y} \leq \frac{1}{x}$$

$$\text{① } \frac{1}{x} \leq \frac{1}{y} \leq \frac{1}{x}$$

$$\downarrow x = 10^6$$

$$\frac{1}{x} \leq \frac{1}{y} \leq \frac{1}{x}$$

$$(18)$$

$$\frac{1}{x} \leq \frac{1}{y} \leq \frac{1}{x}$$

$$\frac{1}{x} \leq \frac{1}{y} \leq \frac{1}{x}$$

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$K[x]$, K -field.

if x does not have a inverse

in $K[x]$.

$\Rightarrow K[x]$ is not a field.

I - maximal ideal

claim: $I = (x)$

$f \in I \Rightarrow f$ is not a unit

if f is unit, then

$$f f^{-1} \in I$$

$$1 \in I$$

$$\Rightarrow I = R$$

$$f = a_1 x + a_2 x^2 + \dots$$

$$f = x(a_1 + a_2 x + \dots)$$

$$f = x g$$

$$\Rightarrow f \in \underline{(x)}.$$

$$\Rightarrow I \subseteq (x) \neq R$$

this is proper ideal of R

but I is maximal ideal

$$\Rightarrow \boxed{I = (x)}$$

these rings are called local rings

$$\pi_4 = \frac{Z_1}{4Z_1} \text{ ideals } \frac{Z_1}{4Z_1} > \left(\frac{2Z_1}{4Z_1} \right), \frac{4Z_1}{4Z_1} = \{0\} \text{ not maximal ideal}$$

ideals of R/I is of the form J/I s.t. $I \subseteq J$

Since Z_{14} is not a field then $\{0\}$ cannot be maximal ideal.

$\frac{2Z_1}{4Z_1}$ is maximal ideal by 3rd

isomorphism thm.

Ques 13) $\mathbb{R} \quad (3) \quad \begin{matrix} \mathbb{R} & \xrightarrow{1} & \mathbb{Q} \\ x & \rightarrow & x \end{matrix}$

0 is maximal in $\mathbb{Q}$

$$\pi^{-1}(0) = 0 \text{ but}$$

not maximal ideal

in $\mathbb{R}$.

$\Rightarrow f$ is continuous map on $\mathbb{R}$
 $\overline{3} > |f(x) - f(y)| < \overline{3}$

$(2) \overline{3} > |f(x) - f(y)| < \overline{3} \Rightarrow$

$f(x) > f(y) > (x-y) > (y-x) \Rightarrow$

$\oplus \text{ by } x > y-x > y-x \Rightarrow$

$x > |y-x|$

$x > 3 = (y)$

$\Rightarrow 3 \leq x \leq 3$

$\Rightarrow f$ is onto.

as f is continuous

$\oplus \text{ — } f(a) \geq f(b) \Rightarrow$

$f(a) - f(b) \geq 0$

$f(a) - f(b) = (a-b)^2$

$f(a-b) = (a-b)^2$

$a-b = x^2, x \in \mathbb{R}$

$a \geq b \Rightarrow f(a) \geq f(b)$
 continuous in $\mathbb{R} \rightarrow \mathbb{R}$

$\overline{3} > |f(x) - f(y)| < \overline{3}$

$f(x) > f(y) > (x-y) > (y-x)$

relation:

...?

is its in $\mathbb{R}$ and $\{ \}$ is non-empty

$$f(0) = 0 \quad \forall x \in \mathbb{R}$$

$$f(1) = x = 1$$

$$\Rightarrow \boxed{x=1}$$

$\Rightarrow f$ is identity

$\Rightarrow$ mp is a maximal ideal

⑥ if $\mathfrak{p}$ is compact, any maximal ideal $\mathfrak{m}$

of this type. let $\mathfrak{p}$ be a maximal ideal of $C(X)$

if $\mathfrak{p} \neq \mathfrak{m}$ for some $\mathfrak{m} \in X$

as $\mathfrak{p}$ is maximal

$$\Rightarrow \mathfrak{p} = \mathfrak{m}$$

suppose $\mathfrak{p} \neq \mathfrak{m}$ $\forall \mathfrak{m} \in X$

$$\forall \mathfrak{m} \in X : \exists f \in \mathfrak{p} \text{ s.t. } f(\mathfrak{m}) \neq 0$$

$$f \cdot x \rightarrow \mathfrak{p} \text{ as } x \text{ and } f(\mathfrak{m}) \neq 0$$

$\Rightarrow$ $\mathfrak{p}$ is some open set $U_{\mathfrak{p}}$ containing

$$\mathfrak{p} \text{ s.t. } \mathfrak{p} \neq 0 \text{ on } \overline{U_{\mathfrak{p}}}$$

$$\{U_{\mathfrak{p}} \mid \mathfrak{p} \in X\} \text{ covers } X$$

but X is compact

$$\exists \mathfrak{p}_1, \mathfrak{p}_2, \dots, \mathfrak{p}_n \text{ s.t.}$$

$$X = \bigcup_{i=1}^n U_{\mathfrak{p}_i}$$

$$\mathfrak{p} = \mathfrak{p}_1^2 + \mathfrak{p}_2^2 + \dots + \mathfrak{p}_n^2 \subseteq \mathfrak{p}$$

$$\text{and } \mathfrak{p} \neq 0 \text{ on } X \Rightarrow \frac{\mathfrak{p}}{1} \cdot X \rightarrow \mathfrak{p} \text{ is}$$

maximal.

$$\Rightarrow \mathfrak{p} = C(X)$$

①

$$f(0) = 0$$

$$B(p/q) = p/q$$

if $f(x_0) + x_0$ has same x_0

$$f(x_0) > x_0$$

my continuity problem

$$x_0 < x_1$$

$$x_0 < f(x_0) < x_1$$

$$x_0 < f$$

$$f(x_0) < f(x_1) = x_1$$

$$x_0 < x_1$$

$$\Rightarrow f(x_0) < f < f(x_1)$$

$$= f$$

$$f(x) = x \quad \forall x \in \mathbb{R}$$

hence

case (ii)

$$m_p = \{ f(x) \in C(x) \mid f(p) = 0 \}$$

$$C(x) \xrightarrow{\phi} \mathbb{R}$$

$$f \rightarrow f(p)$$

then ϕ is surjective and onto

$$\frac{C(x)}{Ker \phi} \cong \mathbb{R}$$

$$Ker \phi = m_p$$

$$\Rightarrow \text{since } \mathbb{R} \text{ is field } \frac{C(x)}{C(x)} \text{ is field}$$

Ques (7) $\mathbb{Z}[X]$ is not a PID.

Suppose R is CRU

then $\frac{R[X]}{\langle X \rangle} \cong R$

[proof:

$$\phi: R[X] \rightarrow R$$

$$\text{by } \phi(f(x)) = f(0)$$

then clearly ϕ is ring homo. and onto.

then by first isomorphism thm,

$$\frac{R[X]}{\text{ker } \phi} \cong R, \text{ as } \text{ker } \phi = \langle X \rangle$$

$$\Rightarrow \boxed{\frac{R[X]}{\langle X \rangle} \cong R}$$

$$\Rightarrow \frac{\mathbb{Z}[X]}{\langle X \rangle} \cong \mathbb{Z}$$

if $\mathbb{Z}$ is PID, then Thm [in PID, every non zero prime ideal is a maximal ideal]
to show $\langle X \rangle$ is a non-zero prime ideal
of $\mathbb{Z}[X]$. clearly as $f(x) = x \notin \langle X \rangle$

$$\Rightarrow \langle X \rangle \neq 0$$

and let $f(x)g(x) \in \langle X \rangle$ then

$$f(x)g(x) = xh(x), \text{ for some } h(x) \in \mathbb{Z}[X]$$

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 solution: $\rightarrow$
 $R \times S$ is $I \times J$ where $I \subset R$ & $J \subset S$
 $K \subset R \times S$
 $I = aSR$
 $I \times J$ ideal ideal

as $\mathbb{Z}$ is integral domain
 $\Rightarrow \langle x \rangle$ is prime ideal in $\mathbb{Z}[x]$.
 $\Rightarrow \langle x \rangle$ is maximal ideal in $\mathbb{Z}[x]$.
 $\Rightarrow \frac{\mathbb{Z}[x]}{\langle x \rangle} \cong \mathbb{Z}$ is field which is
 a contradiction as $\mathbb{Z}$ is not a field.

To show in PID, any non zero prime
 ideal is a maximal ideal.

proof: let R - PID
 and $0 \neq P$ - prime in R

then $\exists a \in R$ s.t. $P = \langle a \rangle$

suppose $P \subset I \subset R$ for some ideal

To show: $P = I$ or $I = R$.

let $I = \langle b \rangle$, $b \neq 0$ for some $b \in R$

as $P \subset I$, $\Rightarrow a = bu$ for some $u \in R$.

as $a \in P \Rightarrow bu \in P$

$\Rightarrow bu \in P$ or $u \in P$

if $b \in P$, then $I \subseteq P$

$$\Rightarrow I = P$$

if $a \in P$, $u = as$ for some $s \in R$

$$bu = bas$$

$$a = bas$$

$$a(1-bs) = 0$$

Since R is an integral domain always &
 $a \neq 0$.

$$\Rightarrow 1 = bs$$

$\Rightarrow b$ is invertible

$$\Rightarrow I = R$$

$\Rightarrow P$ is maximal ideal in R

this proves that $\mathbb{Z}[x]$ is not a PID.

Que (3) Given

Solution $\rightarrow$

①

Let K ideal of $R[x]$, let define

$$I = \{a \in R : \exists b \in S \text{ s.t. } (a,b) \in K\}$$

$$J = \{b \in S : \exists a \in R \text{ s.t. } (a,b) \in K\}$$

① I ideal of R

let $x, y \in I$, then

$$\exists a, b \in S: x = a, y = b$$

$$(x, b_1), (y, b_2) \in K$$

since K is ideal

$$\Rightarrow (x - y, b_1 - b_2) \in K$$

$$\Rightarrow x - y \in I$$

and let $x \in I, \mu \in R$
 $\exists a \in S: x = a$
 $(x, a) \in K$ and $(\mu, 1) \in R \times S$

$$(\mu x, \mu a) \in K$$

$$\Rightarrow (\mu x, a) \in K$$

$$\Rightarrow \mu x \in I$$

$$\Rightarrow I \text{ ideal of } R$$

② $K \subseteq I \times J$, since let $(a, b) \in K$.
 $\Rightarrow a \in I, b \in J$

$$\Rightarrow (a, b) \in I \times J$$

$$\Rightarrow K \subseteq I \times J$$

let $(a, b) \in I \times J$

$$\Rightarrow a \in I \text{ and } b \in J$$

$$\Rightarrow \exists b_1 \in S \text{ s.t. } (a, b_1) \in K$$

$$\exists b_2 \in P \text{ s.t. } (b_2, b) \in K$$

as K ideal of $R \times S$

$$\Rightarrow (1, 0)(a, b_1) = (a, 0) \in K$$

$$(0, 1)(b_2, b) = (0, b) \in K$$

$$\Rightarrow (a, 0) + (0, b) \in K$$

$$\Rightarrow (a, b) \in K$$

$$I \Rightarrow I \times J \subseteq K$$

$$\Rightarrow \boxed{I \times J = K}$$

now, let $I \times J$ prime ideal of $R \times S$.

$\frac{R \times S}{I \times J}$ is a domain

$$\text{as } \frac{R \times S}{I \times J} \cong \frac{R}{I} \times \frac{S}{J} \text{ — domain.}$$

as if $A \times B$ is domain then

either A is domain and $B = 0$ or

B is domain and $A = 0$

i.e. I is prime and $J=S$ on

$I=R$ and J is prime.

So, (only) prime ideals are

$P \times S$ where P is prime ^{ideal} on

$R \times T$ where T is prime ideal.

$$\text{to } \frac{R \times S}{I \times J} \cong \frac{R}{I} \times \frac{S}{J}$$

$$\phi : \frac{R \times S}{I \times J} \rightarrow \frac{R}{I} \times \frac{S}{J}$$

$$(a, b) \rightarrow (a+I, b+J)$$

then clearly ϕ is an isomorphism

$$\boxed{\frac{R \times S}{I \times J} \cong \frac{R}{I} \times \frac{S}{J}} \quad (*)$$